

COM3502-4502-6502 SPEECH PROCESSING

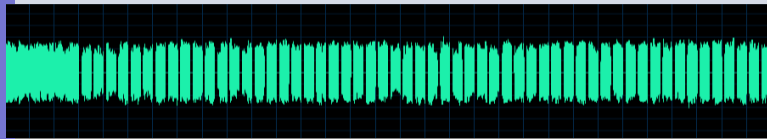
Lecture 11

Hearing

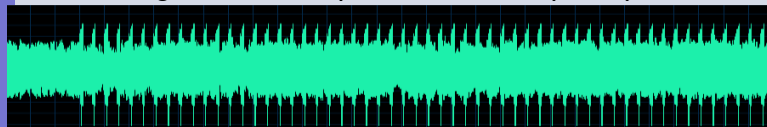


Initial Example: Pulsation threshold

Signal: music + pauses (125/500 ms)

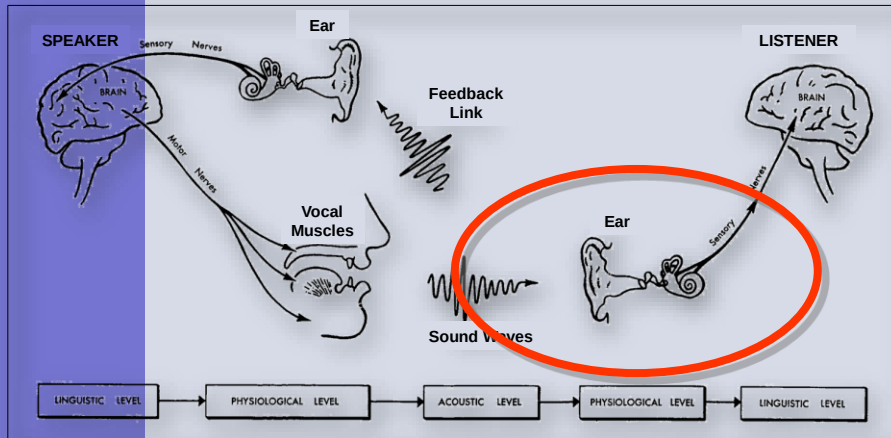


Signal: music + pauses + noise (+6dB)



- Continuity of perception
 - Perceptive analysis: extrapolation of dynamics
 - Redundancy of the source

The Speech Chain

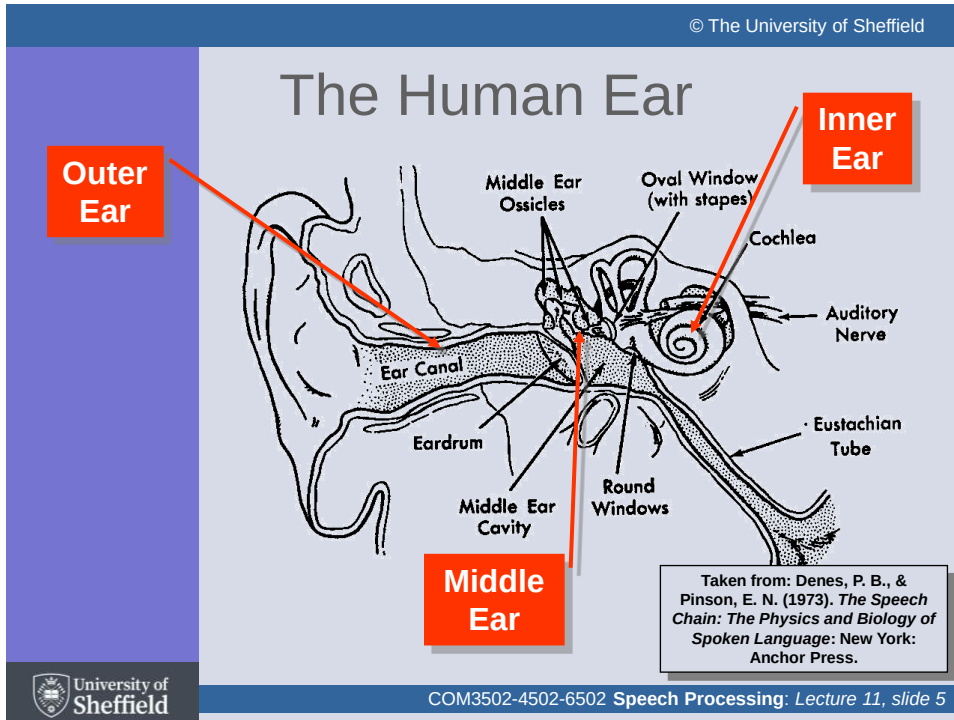


Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*. New York: Anchor Press.

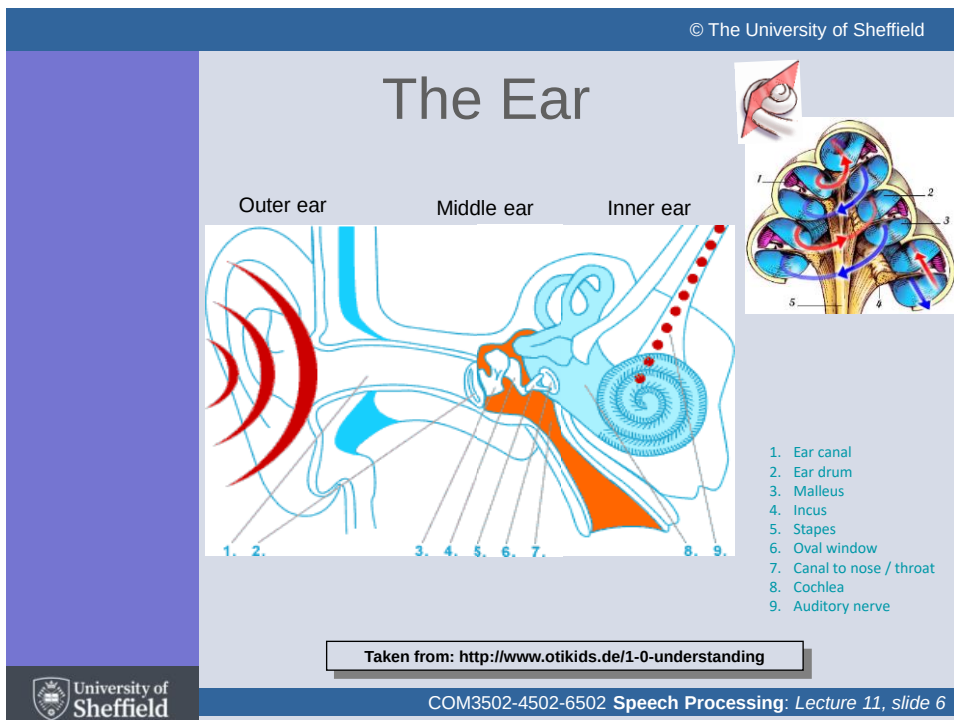
The Human Ear



- The auditory system has evolved for ...
 - acoustic sensing
 - sound localisation
 - communication
- The ear is more *general purpose* than the articulatory system
- Its main function is **frequency analysis**
- The main percepts are ...
 - pitch
 - loudness
 - timbre

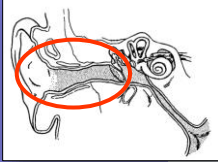


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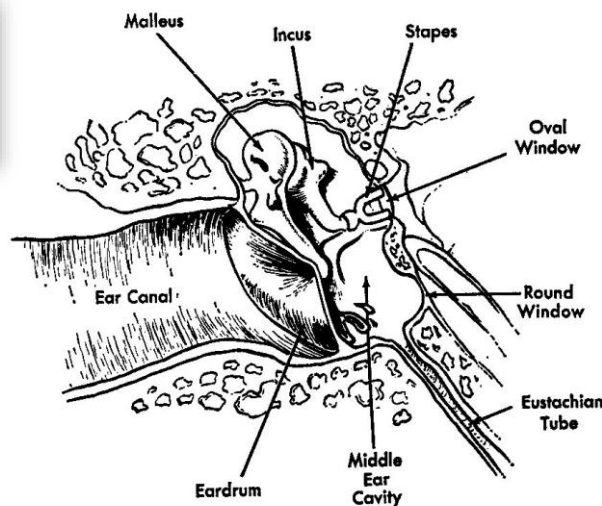
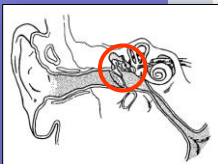
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The Outer Ear



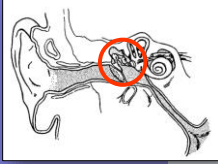
- The '**pinna**' protects the entrance to the ear canal, and its shape makes it directionally sensitive at high frequencies
- The external canal - '**meatus**' - is a tube (~2.7 cm long, ~0.7 cm in diameter) that leads from the pinna to the middle ear
- The meatus terminates at the cone shaped '**tympanic membrane**' (eardrum)
- Sound waves entering the ear impinge upon the eardrum and cause it to **vibrate**

The Middle Ear



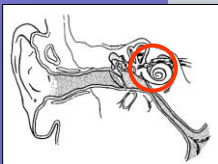
Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*. New York: Anchor Press.

The Middle Ear

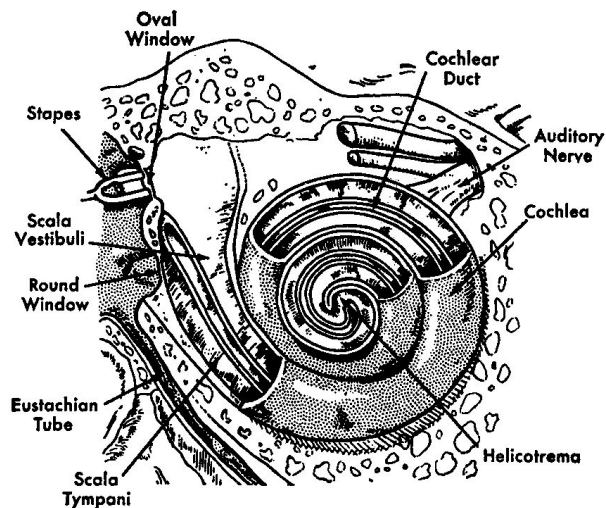


- The middle ear transforms the vibration of the eardrum into oscillations of the liquid in the inner ear by vibrating the '**oval window**'
- The necessary impedance matching (*between air and liquid*) is achieved by a group of bones - the '**ossicles**' - acting as a system of mechanical levers
- The pressure at the oval window is ~35x greater than that arriving at the eardrum
- This mechanical amplification allows us to hear sounds 1000x weaker than otherwise
- Muscles attached to the ossicles protect the inner ear from potential damage due to high sound levels

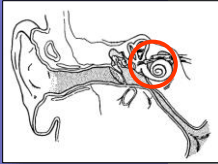
The Inner Ear



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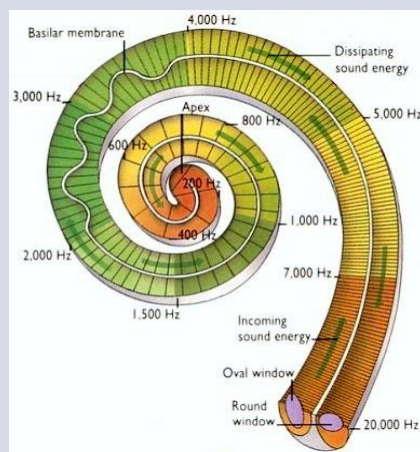
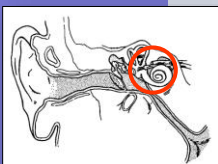


The Inner Ear

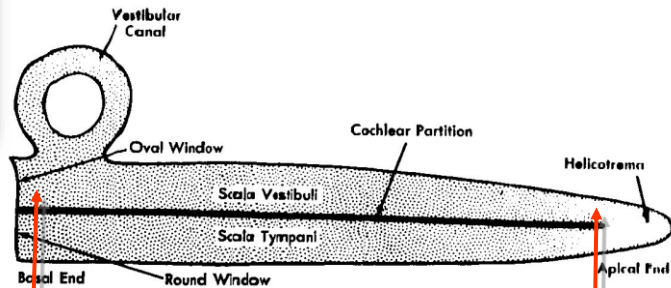
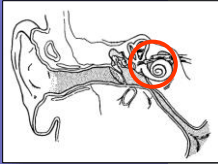


- The transformation from mechanical vibrations to electrical nerve impulses (*'neural transduction'*) takes place in the snail-like structure of the *'cochlea'*
- The cochlea is ~35 mm long and is filled with a colourless liquid called *'perilymph'*
- The cochlea is divided into two regions along its length by a membrane structure called the *'cochlea partition'* (a channel filled with a liquid called *'endolymph'*)
- The cochlea partition is bounded by ...
 - the *'basilar membrane'*
 - *'Reissner's membrane'*

Frequency Analysis in the Cochlear



The Cochlea (*unwound*)

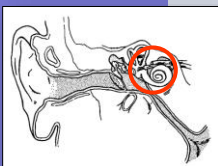


Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*. New York: Anchor Press.

High
Frequencies

Low
Frequencies

Action of the Cochlea

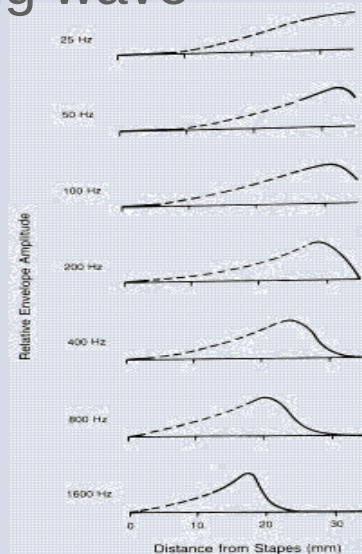


- The mechanical properties of the basilar membrane determine how the cochlea responds to sound
- Vibrations entering at the oval window set up **travelling waves** which lead to peaks of energy at different places along the cochlea depending on the frequency
- The vibration is nearest the oval window for high-frequency sounds
- The '**organ of corti**' transform the mechanical movements into electrochemical pulses by bending the outer hair cells (*of which there are ~25,000*)
- These actions are equivalent to a bank of '**bandpass filters**'

Traveling wave



- Different frequencies of sound excite maximum vibrations at different places of basilar membrane.
- Frequency of vibration at a given place is equal to that of the nearest stimulus component (resonance).
- Cochlea performs a frequency analysis.



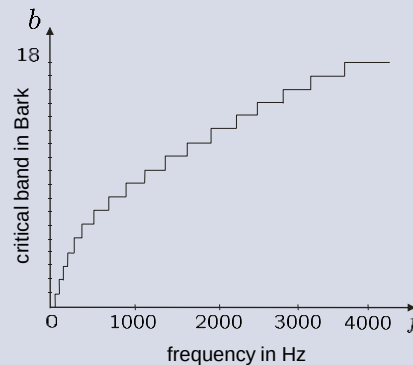
Spectral Resolution of the Human Auditory System



Frequency-to-Bark-Transform

$$b = 13 \cdot \arctan\left(0.76 \frac{f}{1000}\right) + 3.5 \cdot \arctan\left(\frac{f}{7500}\right)^2$$

Bark	f_{low} (in Hz)	f_{high} (in Hz)
0	0	100
1	100	200
2	200	300
3	300	400
4	400	510
5	510	630
6	630	770
7	770	920
8	920	1080
9	1080	1265
10	1265	1480
11	1480	1715
12	1715	1990
13	1990	2310
14	2310	2690
15	2690	3125
16	3125	3675
17	3675	4350



Heinrich Georg Barkhausen (*1881 in Bremen)

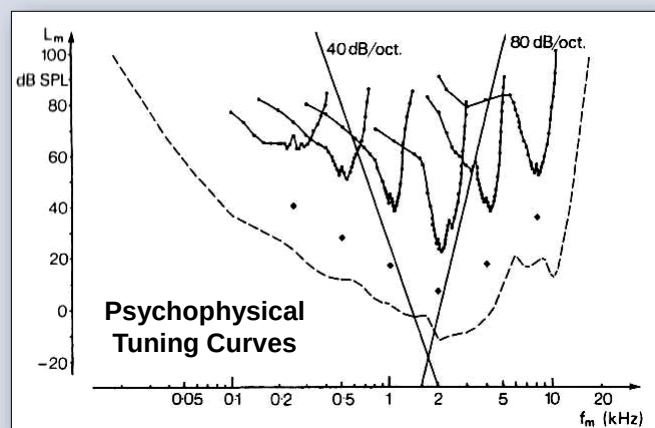
1 Bark = 100 Mel

Frequency Selectivity



- The frequency range of human hearing lies between ~20 Hz and ~20,000 Hz
- The ability to discriminate energy varies according to frequency ...
 - 0.2 Hz at 100 Hz
 - 1.5 Hz at 2000 Hz
 - 30 Hz at 12,000 Hz
- Low frequency sounds can mask higher frequency sounds because of the overlap between 'auditory filters'
- The bandwidth over which masking operates is termed the 'critical band'
- The shapes of the auditory filters are revealed by deriving 'psychophysical tuning curves'

Frequency Selectivity

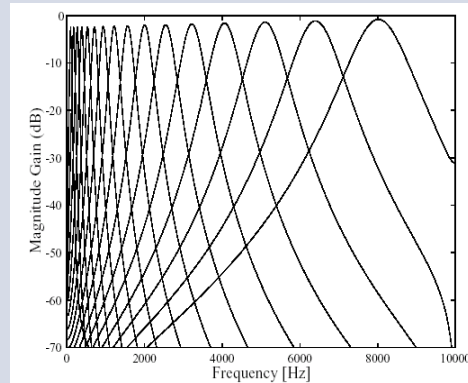


Taken from: Holmes, J. N., & Holmes, W. (2002). *Speech Synthesis and Recognition*: Taylor & Francis.

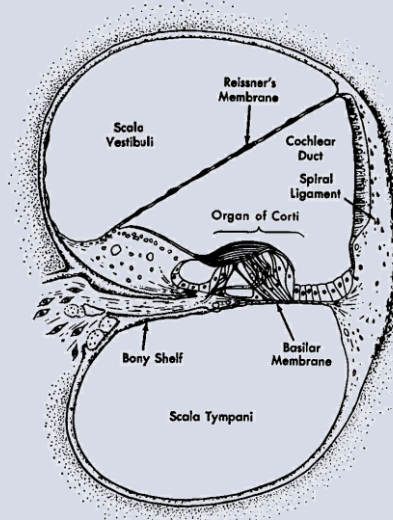
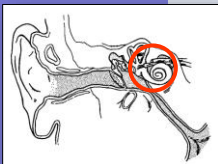
Gammatone filterbank



- Each position on the basilar membrane simulated by a gammatone filter with appropriate centre frequency and bandwidth.
- Smaller number of filters as for spectrograms
- variation in bandwidth with frequency (unlike Fourier analysis).

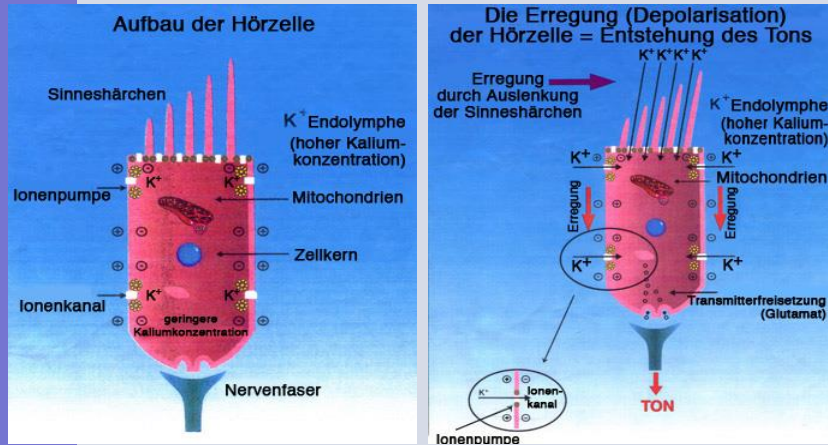


Cochlear Cross-Section



Taken from: Denes, P. B., & Pinson, E. N. (1973). *The Speech Chain: The Physics and Biology of Spoken Language*. New York: Anchor Press.

Hearing Cells

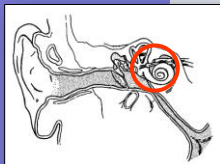


Source: http://dasgesundehoer.de/ohr/208_arbeitet_die_Horzell.shtml

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Dancing Hair Cell

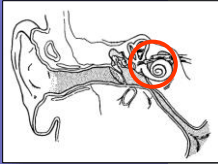


Source: Yale School of Medicine

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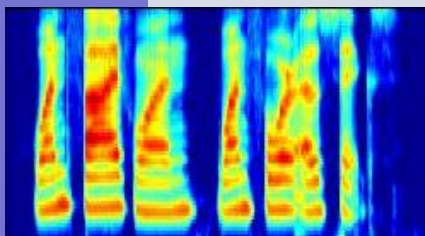
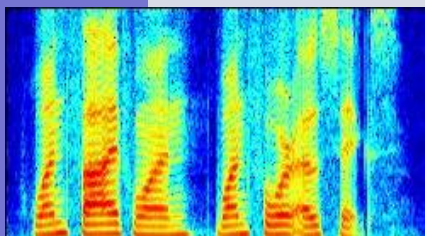
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Action of the Cochlea



<https://youtu.be/dyenMluFaUw>

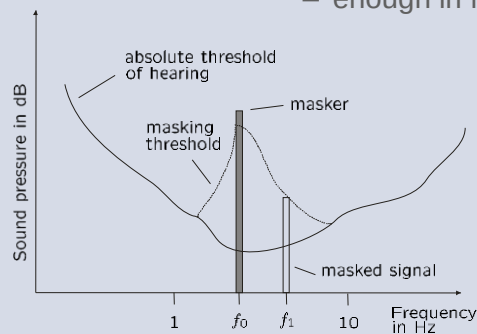
Spectrogram vs. Cochleogram



- Spectrogram
 - Plot of log energy across time and frequency (linear frequency scale)
- ‘Cochleogram’
 - Cochlear filtering by the gammatone filterbank (or other models of cochlear filtering)
 - Quasi-logarithmic frequency scale, and filter bandwidth is frequency-dependent

Masking

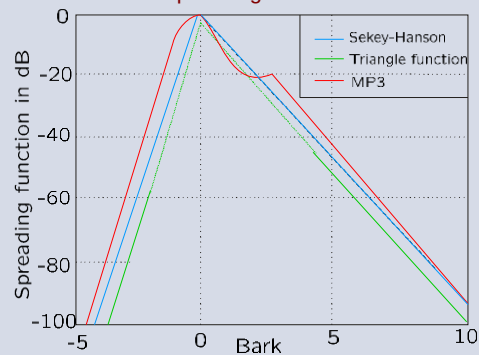
- (Frequency) Masking effect:
 - the weaker signal can be made inaudible by the stronger signal,
 - i.e., the masker, if they are close
 - enough in frequency and time.



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Spreading Function

Three approximations for the spreading function:

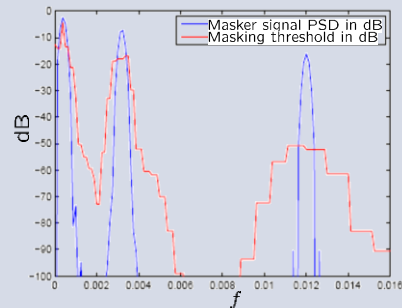


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Masking Threshold

Example:

$$s_n(t) = \sin(2\pi 100t) + 0.8\sin(2\pi 800t) + 0.5\sin(2\pi 3000t)$$



The masking threshold can be calculated from the clean speech signal :

$$\Phi_{TT}[m, l] = f(\Phi_{s_n s_n}[m, l])$$

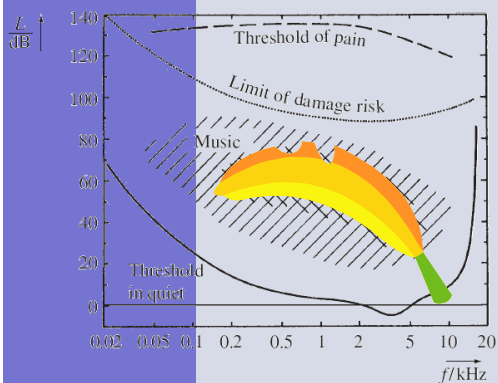
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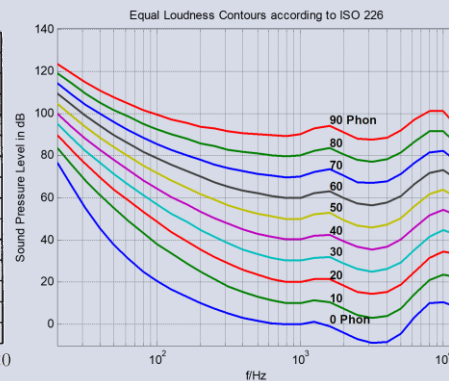
Hearing Area / Equal Loudness Curves

The Hearing Area



[Zwicker, Fastl 1999]

Equal Loudness Curves



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Frequency Range of Auditory System

- Frequency range: 20 Hz - 20 kHz
- Intensity range: between 10^{-16} Watt/cm² and 10^{-2} Watt/cm², or using logarithmic presentation → between 0 dB and 140 dB.
- Hearing range of some (other) animals:
 - Frog: 50-10000Hz
 - Dog: 15-50000Hz
 - Cat: 60-65000Hz
 - Bats: 1000-120000Hz

Question ...

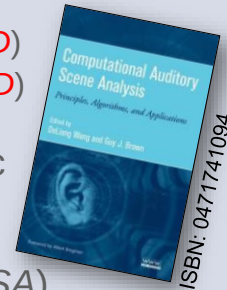
What's the point of having two ears?



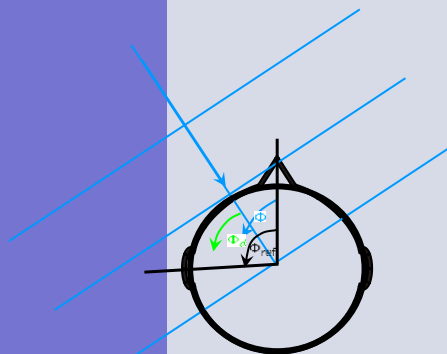
Binaural Auditory Processing



- Sound localisation
 - to direct (*visual*) attention
- Possible mechanism:
 - inter-aural time differences (*ITD*)
 - inter-aural level differences (*ILD*)
- Listening in complex acoustic environments
- ‘Auditory Scene Analysis’ (ASA)



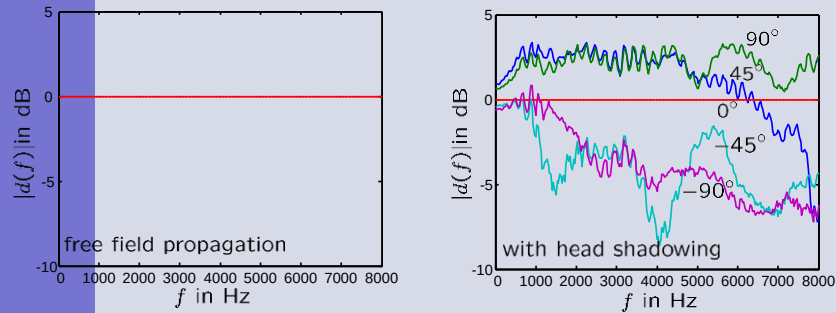
DOA Estimation at the human head



Head-related binaural cues for DOA estimation:

- Interaural Time Difference (ITD)
wave has to travel longer way
“around the head”
- Interaural Level Difference (ILD)
Shadowing / reflection of parts of the
wave front

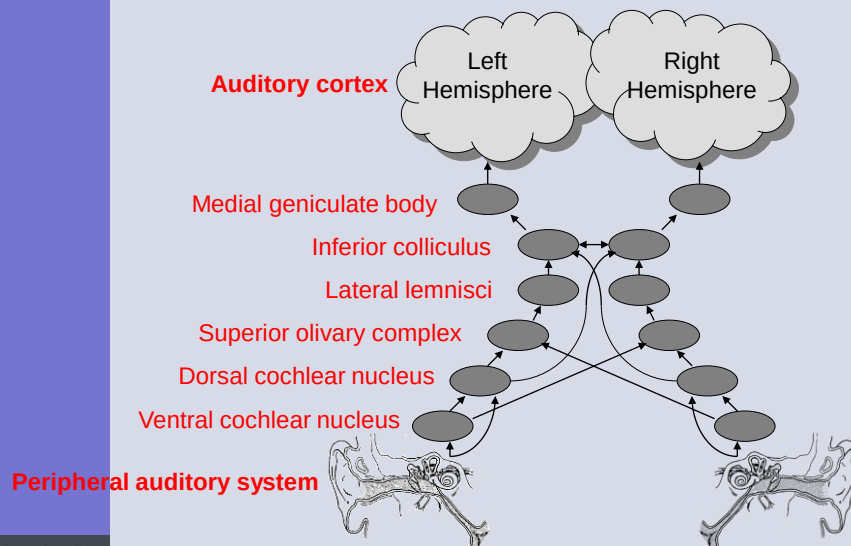
Free-field and head-related wave propagation



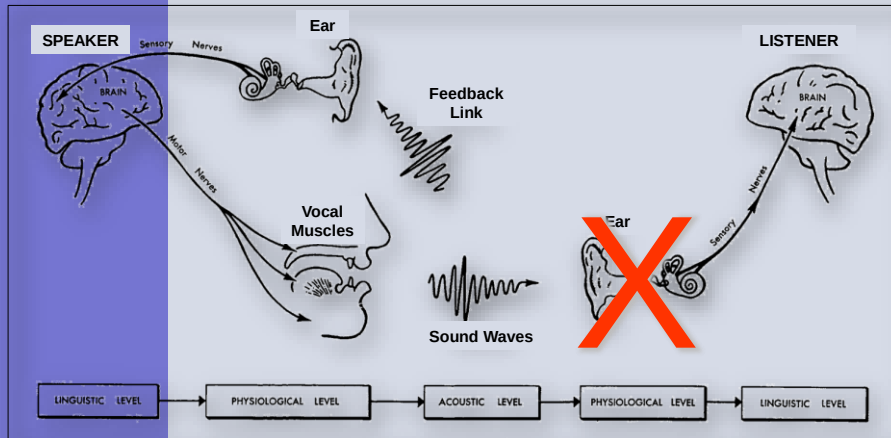
Head-related wave propagation is user-specific (depends on the pinnae, head size, etc.)

Head shadowing effects

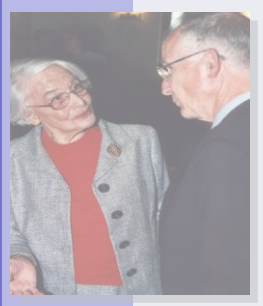
Auditory Pathways



Hearing Impairment



Numbers: Hearing Impairment



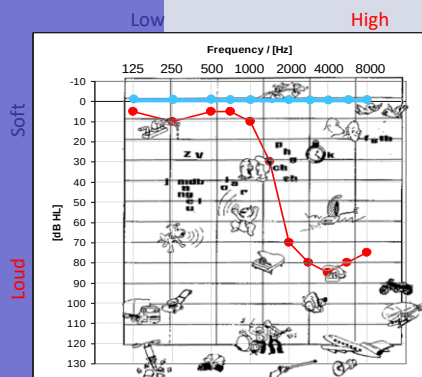
- 16% of Europeans (approx. 70 Mio.) feel affected by hearing loss in everyday life
- About 40% of 61-70 years old and more than 65% of above-70 years old are hearing loss candidates
- Hearing loss becomes the most common sensory disability
- In 2015: 700 Mio. people worldwide suffer from hearing loss, 2025 over 900 Mio.

How to Recognize Hearing Loss?



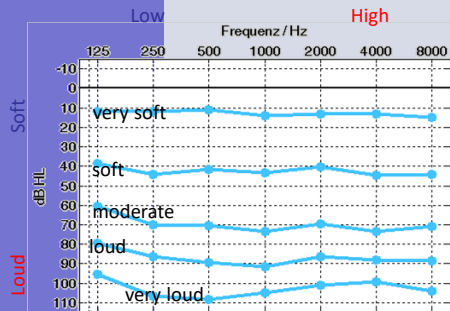
- Birds and falling leaves are not perceived any more
- TV is too loud
- Often inquiry
- Often words are not understood correctly
- No longer involved in discussions in a group
- Increased listening effort leads to tiredness in the evening

What is hearing loss?



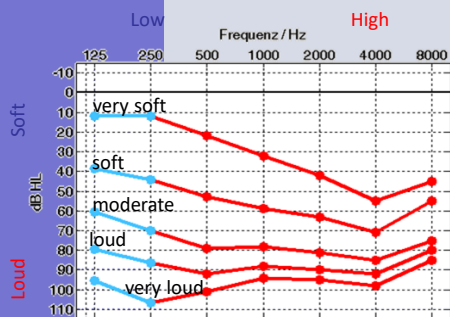
- Soft sounds are difficult to hear clearly
- Hearing threshold in frequency increases
- Loud sounds become too loud
- Dynamic range in frequency is reduced
- Speech in noise is no longer understood correctly
- Psychoacoustic benefits is declined (e.g. Frequency / Time resolution)

What is hearing loss?



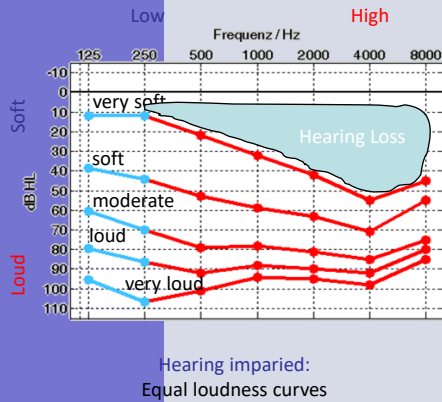
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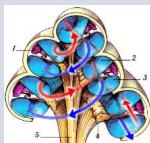


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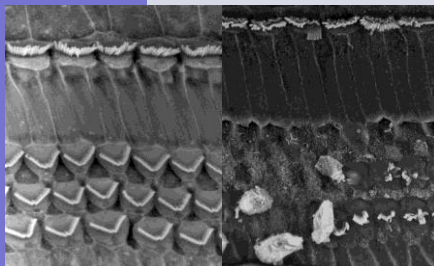
What is hearing loss?



Intact hair cell



Damaged hair cell



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Consequences of hearing loss

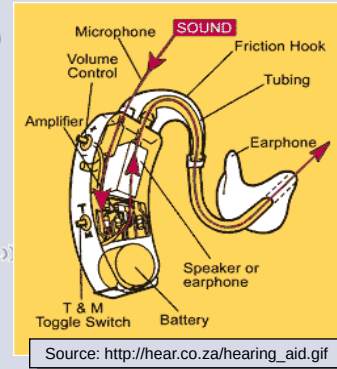
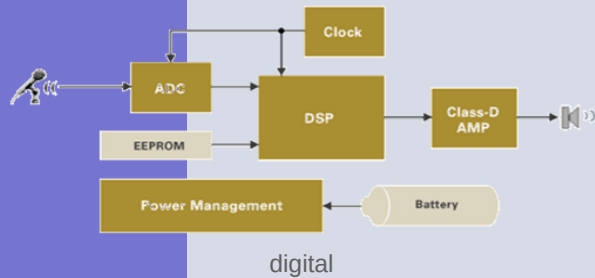
- Communication may be difficult to follow
- Uncertainty in everyday life
 - Frustration and shame
- Withdrawal, due to
 - missing what others are taking about
 - not constantly ask
 - everything is hard
- May lead to loneliness
- "pro-active strategy" responses to what people think they understood
- **Danger:** confusion dementia with hearing loss

Hearing Aids



Basic-Components of hearing aids

- Microphone
- Amplifier (in digital also DSP)
- Loudspeaker
- Battery



Source: http://www.sicologic.com/11hearing_aid/jpg/medical_hearing_aid.gif

Types of hearing aids



BTE - behind the ear



Mini BTE - behind the ear



ITE - in the ear



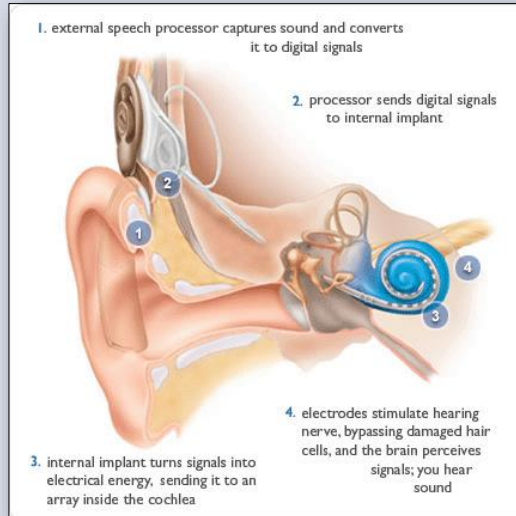
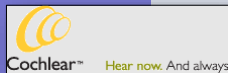
ITC - in the canal



CIC - completely in the canal

[Source: http://www.betterhearing.org/pdfs/e-Guides/Your_Guide_to_Hearing_Aids.pdf]

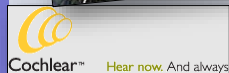
Cochlear Implants



Cochlear Implants



<http://www.actiononhearingloss.org.uk>



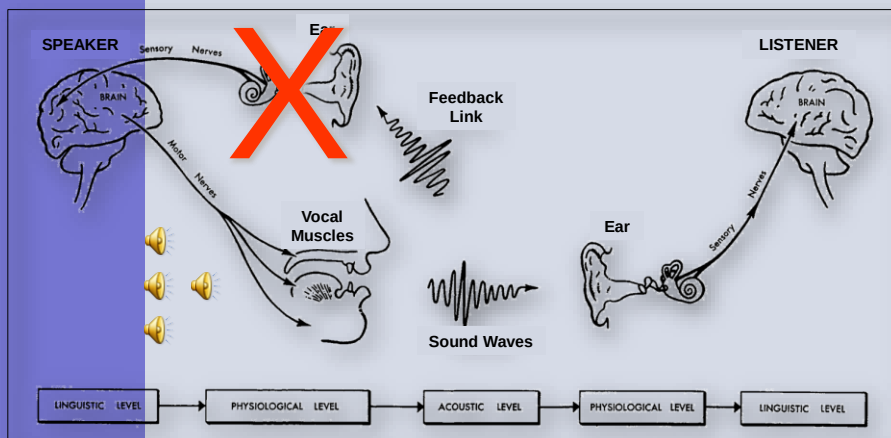
<https://youtu.be/HTzTt1VnHRM>

The electrode array



Source: <http://www.earsurgery.org>

Speaking Also Depends on Hearing



This slide set has covered ...

- Outer, middle and inner ear
- Action of the cochlea
- Spectrographic analysis
- Binaural processing
- Auditory psychophysics
- Hearing impairment

Any Questions ?



Next time ...

The Nature of Speech